

CCS6224

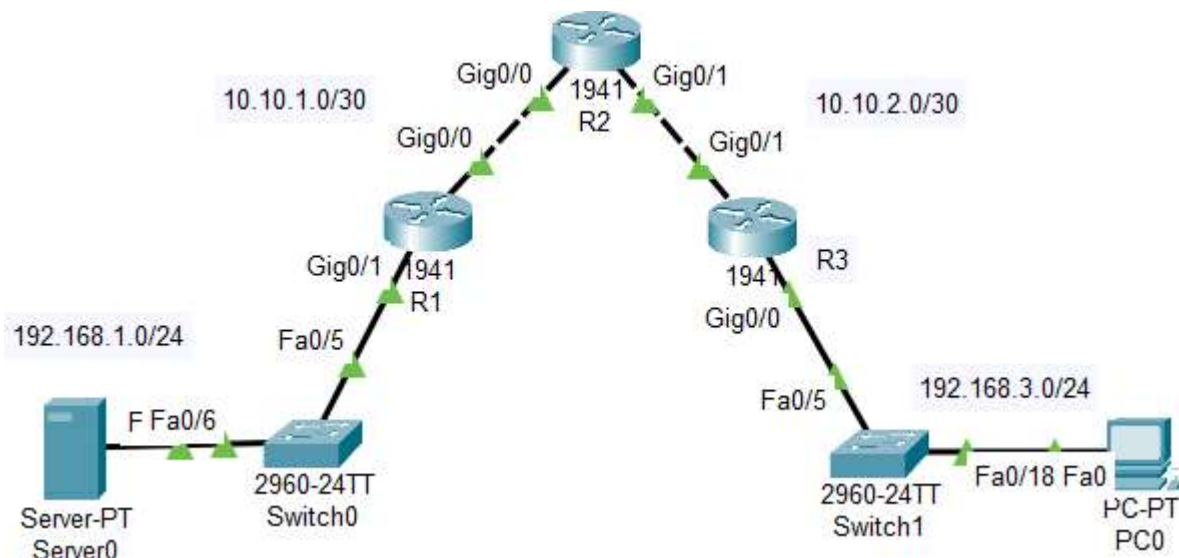
NETWORK SECURITY

Lab 2

(Graded Lab Exercise 1)

Practical Lab: Security Configuration of Router for Administrative Access

1. Construct the following network topology using the Packet Tracer.



Note: For graded lab exercise submission, students are required to use the following IP addresses instead:

192.168.1.0/24 → 192.StudentDayOfBirth.StudentMonthOfBirth.0/24

192.168.3.0/24 → 192.StudentDayOfBirth.(StudentMonthOfBirth+1).0/24

10.10.1.0/30 => 10.StudentDayOfBirth.(StudentMonthOfBirth).0/30

10.10.2.0/30 => 10.StudentDayOfBirth.(StudentMonthOfBirth+1).0/30

Failure to comply with this instruction will result in mark reduction. The same goes for 3.a and 3.c below.

2. Configure the network devices with the following basic configurations. **(2 marks)**

- a) IP address
- b) Routing protocol RIP v2

3. Perform the administrative access controls for router R1.

a) Configure and encrypt all passwords. **(2 marks)**

- Set password to min. length = 8
- Set line to log out after 3 minutes of inactivity
- Set enable secret *cisco123*
- Set console password *ciscoconpw* (use *StudentIDconpw* for lab exercise submission)
- Set telnet password *ciscovtypw* (use *StudentIDvtypw* for lab exercise submission)
- Encrypt all clear text passwords in running-config file

Note: StudentID is your ID given by MMU.

b) Configure the login warning banner "*This router is strictly for authorized access only!*"

c) Create a new user account *userA* (use *StudentID* for lab exercise submission), with a secret password *userApass* (use *StudentIDpass* for lab exercise submission). **(2 marks)**

- Set the console line to use the local user account, *userA*
- Type "exit" to log out from the router, use the *userA* to login
- Test whether or not *userA* can view the router setting in running-config file
- Set the vty lines to use the same local user account, *userA*
- Telnet from PC0 to the router

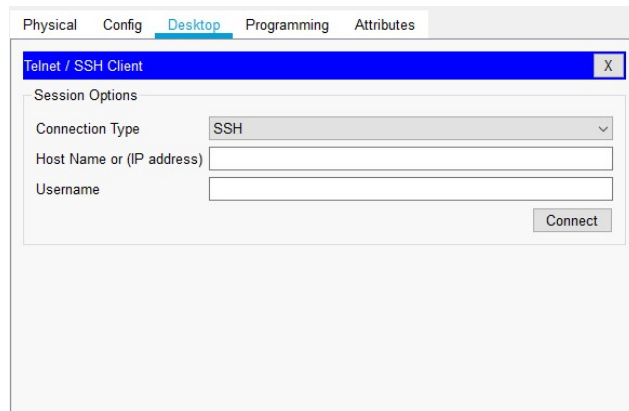
Note: StudentID is your ID given by MMU.

d) Configure SSH server on router R1 **(2 marks)**

- Set *fci_network.com* as the domain name in R1
- Configure a user *admin* with the highest privilege level, set password *ciscoadmin*
- Configure the vty lines to only accept SSH connections
- Erase existing key pairs on the router
- Create the RSA encryption key pair for the router, with 1024 for the number of modulus bits
- Set the IP SSH to version 2, verify the SSH configuration
- Set SSH timeout to 1 minute and retries to 2

e) Verify the connectivity of an SSH client from PC0

- At PC0, open the SSH client application



- Input the IP address of R1 in the Host Name field
- Input the username *admin* and then press connect
- Input the password of *admin*
- If the SSH login is successful, you will be in the privileged exec mode of R1
- Type "show users" to check how many users are accessing R1 at the moment
- Close the SSH client application. Try to use command prompt to connect R1 using telnet, successful or not?

f) Configure SCP server on router R1 **(2 marks)**

Since SSH is configured on R1, so we can securely transfer files by using the SCP protocol

- First, enable AAA on the router
- For AAA authentication, set the local database as the default login authentication method
- For AAA authorization, set the local database as the default command authorization
- Enable SCP server on R1

Test the SCP service, by transferring the running configuration file from R1 to R3.

- At router R1, save the running configuration on R1 to a file named *R1-config* to flash
- At router R3, use SCP command to pull the configuration file from R1
- Verify that the file has been copied and stored to R3's flash

4. Save all your configuration in a pkt file. The same file will be used for the next lab.

Note: For graded lab exercise submission, follow the instructions posted on eBwise on this matter.